

CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · FORMULA BANK

# Chapter 12: Reflection

Every formula and result of the chapter on one sheet

|    |                                |                                                                     |
|----|--------------------------------|---------------------------------------------------------------------|
| 1  | Reflection in the x-axis       | $(x, y) \rightarrow (x, -y)$                                        |
| 2  | Reflection in the y-axis       | $(x, y) \rightarrow (-x, y)$                                        |
| 3  | Reflection in the origin       | $(x, y) \rightarrow (-x, -y)$                                       |
| 4  | Reflection in the line $x = a$ | $(x, y) \rightarrow (2a - x, y)$                                    |
| 5  | Reflection in the line $y = b$ | $(x, y) \rightarrow (x, 2b - y)$                                    |
| 6  | Mirror line                    | Perpendicular bisector of the segment joining a point and its image |
| 7  | Invariant point                | A point on the mirror line; its image is itself                     |
| 8  | $M_x$ followed by $M_y$        | Same as reflection in the origin                                    |
| 9  | Equation of the x-axis         | $y = 0$                                                             |
| 10 | Equation of the y-axis         | $x = 0$                                                             |
| 11 | Figure and its image           | Congruent, with lengths and angles unchanged                        |